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Combustor Diffuser Modeling
Part V: Validation with a Three Passage Diffuser Rig Data

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ABSTRACT

A full-scale 3-passage combustor diffusion system has been analyzed with FLUENT® code using its grid generation package, the standard k-ε model and the nonequilibrium wall function. Inlet conditions were calculated using through the rig inlet, the three profilers that provide center-, inner- and outer-peak profiles at the prediffuser inlet. 3-D model predictions were compared with data from a full-scale annular diffuser rig. In general, CFD gave good agreement with data as shown in the following table.

INTRODUCTION

2-D and 3-D CFD calculations are generally done during the gas turbine combustion system design process to aid in the optimization of the diffusion system. Early work by Karki et al[1] as well as the follow-on publications, e.g., Reference 2 have shown that the standard k-ε model with wall functions overpredicts C_P by 5-15% compared to data. A preliminary assessment of various advanced turbulence models indicates[3] that SST k- ω turbulence model has shown potential for giving better agreement with data for flows subject to adverse pressure gradients including separation. However due to lack of systematic validation of CFD-based approach with data of combustor diffusion systems of practical interest to designers, the current design process relies on empirical data collected on simulated full-scale 3-D diffuser rigs coupled with qualitative guidance from CFD.

Prompted by recent advances in CFD simulation technology, turbulence modeling[4, 5] and availability of an excellent data set[6] from a typical single passage full-scale annular diffuser rig, we have undertaken a comprehensive investigation to study the effect of the following on C_P and Ω:

1. Diffusion system design parameters including inlet profiles, pre-diffuser area ratio, L/H, cowling, blockages due to dome swirlers, nozzle stem and tip, airflow splits between the dome, inner and outer annuli of the combustor liner.
2. Procedures for calculating diffuser inlet conditions.
3. 2-D versus 3-D simulations.
4. Different turbulence models and near-wall treatments to capture boundary layer losses and flow separation.

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The Part I paper[6], showed that best agreement with data is obtained with the standard k-ε model starting with the calculated diffuser inlet conditions instead of estimating it from 4-5 P_t rakes and the accompanying wall static pressure taps. With measured diffuser inlet profiles and 2-D simulation, the average overprediction for prediffuser CP is 27, 23.5 and 12.5%, respectively for Runs 3, 4 and 5. However, these overpredictions reduce to 7.5, 6 and -2.5%, respectively with the calculated inlet profiles; negative value means underprediction. The corresponding errors with SST k-Ω model are 12.5, 17 and 2.5%. The average underpredictions for Ω are 12.5, 0 are –2.5%; here negative number means overprediction. 2-D simulation gave very poor agreement with the dome total pressure recovery coefficient data; prediction is on the average 27.5% lower than data.

Part II results[7], as summarized in Table 2, clearly show that 3-D simulations give significantly improved correlation with data than 2-D. Moreover, the turbulence, mean velocity flow field characteristics, total and static pressure distribution are captured more realistically with 3-D simulations.

Part III tried[8] to predict prediffuser separation for the following two cases, and shows that none of the existing turbulence models of FLUENT® get good correlation with data. Fortunately, as long as combustor diffusers are designed conservatively we don't need to worry about predicting flow separation within the prediffuser passage.

Part IV deals[9] with predicting the performance of the single passage diffusion system:

1. With and without dome cowling Run 3 versus Run 6 of Table 1.
2. Increasing nozzle diameter from 1.7" to 1.9" along with minor accompanying changes in the cowl geometry, Run 3 versus Run 11.
3. Increasing nozzle stem blockage by 20% from Run 11, Run 11 versus Run 12.

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Model predicts well for (1), and is more reliable in quantifying the effects mentioned in (2) and (3) than the data from a typical engineering quality diffuser rig.

Part V, the present paper extends 3-D simulation technique of Reference 7 for single passage diffuser to a typical 3-passage full-scale annular diffuser rig, Figure 1. Similar to the single passage study, the standard k-ε model with non-equilibrium wall function[5] is used for making back-to-back comparison between the three prediffuser inlet profiles, namely center-, inner- and outer-peak. The diffuser inlet conditions were calculated by using through-the-diffuser inlet/profiler calculations described in Reference 6. The 3-passage diffuser rig, instrumentation and data sets etc. are similar to those of single passage set-up of References 6 and 7 which also give details on the inlet conditions, turbulence models, wall functions, CFD simulation etc.

CFD COMPARED WITH DATA

CFD calculations and comparison with the data when available are presented in Figures 2 through 14 for the three profilers giving center-, inner- and outer-peak velocity profiles at the diffuser inlet. Figures 2, 3 and 4 present results for the prediffuser and dome with the three inlet profilers, followed in Figures 5 through 10 for the combustor outer and inner passages. Comparison between data (along x-axis) and CFD calculation along y-axis is shown in Figures 11 and 12 Finally Figures 13 and 14 show pictorially the 3-dimensional nature of the diffusion system as indicated by the contours of CP, turbulence kinetic energy and axial velocity along two x-y planes (inline and in-between the fuel nozzles) and the 3.5-r-θ plane through the outer and inner passages. Discussion on these results follows in the following paragraphs.

Prediffuser Exit and Dome CP and Ω

Similar to 3-D calculations for a typical single passage combustor diffusion system[7], here the data and calculations were done for three different prediffuser inlet profiles, namely center-, inner- and outer-peak profiles. Similarly for a given rig set-up, the flow rates through the combustor outer passage, dome and inner passage are controlled to document performance of the diffusion system. Unlike the single passage prediffuser, here one has to estimate the flow splits between the three passages of the prediffuser. It is unlikely that experimentally estimated splits for the prediffuser passages would be identical to those calculated by the CFD simulation. Table 4 compares these splits in addition to listing the range of the flow splits for the combustor dome, outer and inner passages. For example for the center-peak profiler, the experimentally estimated range for inner passage of the prediffuser is between 29.74% and 37.65% compared to CFD calculated range of 31.00% and 34.72%. Consequently when comparing data with CFD for the prediffuser passages in Figure 2, one would usually see different ranges of the flow splits respectively for the data and model calculations.

The diffusion system performance with the center-peak profiler was measured over a wide range of airflow rate through the dome, ranging from ~9%Wa3 to ~37%Wa3, approximately a factor of 4 or a change of 74%. Similarly, airflow splits for the inner and outer passages around the combustor were varied approximately a factor of 2 or changes of 58% and 56%, respectively. However, the resulting airflow split changes for the inner, center and passage of the prediffuser were only 10.7, 3 and 11.2% respectively. Similar observations can be made with the inner- and outer-peak profilers. The airflow split changes in the center passage of the prediffuser of ~3, 3 and 1% are relatively insignificant compared to large changes in the airflow splits for the dome, outer and inner combustor annuli. On the other hand, the airflow split changes in both the outer and inner passages of the prediffuser are approximately comparable with average value being ~9%; which is approximately $1/7^{th}$ of the flow split changes for the dome or $1/5^{th}$ of that of the passages.

Figure 2 shows comparison between data and CFD predictions for the static pressure recovery coefficient CP of the prediffuser exit outer, center and inner passages, respectively going from top to bottom figures; x-axis is the flow split expressed as % total combustor flow, and y-axis value of CP. In general, CFD predictions tend to agree qualitatively except for the center-passage where perhaps data is somewhat more scattered than desirable. As summarized in Table 3 and pictorially presented in Figures 11 and 12, 3-D CFD simulation for the center-peak inlet profiler overpredict CP for the prediffuser exit inner passage by 14.5%, underpredicts for the center passage by ~9% and overpredicts the outer-passage CP by ~17%. However, for the single passage diffuser[7] the corresponding overprediction is only 2.2%, Table 2.The model discrepancy from the data could be reduced by using an advanced turbulence model that accounts appropriately for the boundary layer growth in adverse pressure gradient. It is expected that the shear stress transport k-ω model might be the one to improve prediction capability. However, it continues to be difficult to achieve the desirable level of numerical convergence with SST. Even though experimental accuracy of the total pressure loss coefficient Ω is not as good as in 3-passage diffuser as in the single passage diffuser[7], it is usefully instructive anyway to make comparison with CFD as it calculates Ω correctly.

Table 4: Airflow splits for the prediffuser three passages, combustor dome, outer and inner annuli (passages).

Center-Peak Profiler	Air Flow Splits		
Prediffuser exit:	Max	Min	Change
Inner Passage Exp	37.65%	29.74%	
Inner Passage CFD	34.72%	31.00%	10.7%
Center Passage Exp	40.34%	38.66%	
Center Passage CFD	38.67%	37.51%	3.0%
Outer Passage Exp	29.92%	23.69%	
Outer Passage CFD	30.64%	27.19%	11.2%
Combustor Dome	36.88%	9.53%	74.2%
Comb. Outer Passage	38.43%	17.11%	55.5%
Comb. Iinner Passage	44.30%	18.57%	58.1%

Inner-Peak Profiler	Air Flow Splits		
Prediffuser exit:	Max	Min	Change
Inner Passage Exp	44.71%	37.56%	
Inner Passage CFD	40.99%	38.17%	6.9%
Center Passage Exp	38.80%	35.55%	
Center Passage CFD	37.81%	36.81%	2.7%
Outer Passage Exp	23.64%	19.64%	
Outer Passage CFD	24.29%	22.20%	8.6%
Combustor Dome	37.42%	9.16%	75.5%
Comb. Outer Passage	35.51%	16.06%	54.8%
Comb. Iinner Passage	47.73%	19.78%	58.6%

Outer-Peak Profiler	Air Flow Splits		
Prediffuser exit:	Max	Min	Change
Inner Passage Exp	34.08%	27.67%	
Inner Passage CFD	27.81%	25.57%	8.0%
Center Passage Exp	40.37%	39.35%	
Center Passage CFD	38.17%	37.62%	1.4%
Outer Passage Exp	31.96%	26.57%	
Outer Passage CFD	37.17%	34.04%	8.4%
Combustor Dome	35.69%	9.48%	73.4%
Comb. Outer Passage	39.40%	17.56%	55.4%
Comb. Iinner Passage	43.68%	18.31%	58.1%

Bottom right figure of Figure 2 shows comparison between data and CFD calculated total pressure loss coefficient Ω for the prediffuser exit outer passage. CFD underpredicts Ω by ~29%, Table 3. Similarly CFD underpredicts inner-passage Ω by ~11%. However, CP data for the center-passage is not reliable enough to make any meaningful comparison for its Ω; see also Figure 12.

Figure 2 also shows dome CP and Ω for the center-peak profiler. Interestingly enough, CFD under-predicts dome CP similar to the results for single passage diffuser[7]; underprediction is ~12% versus underprediction for single-passage diffuser of ~8%, Table 2.

Table 5 summarizes CFD-based average values of CP and Ω of the inner, center and outer passages at the prediffuser exit plane, combustor dome, outer and inner passages, respectively, for each of the three profilers, giving center-, inner- and outer-peak profiles. These numbers may be compared with the corresponding average values for a typical single-passage diffuser reported in Reference 7.

Table 5: Average CFD calculated values of CP and Ω of a typical 3-passage diffusion system.

	CFD-based avg.	
Center-Peak Profiler	Cp	Ω
Prediffuser IP exit	0.433	0.203
Prediffuser CP exit	0.346	0.136
Prediffuser OP exit	0.417	0.271
Combustor Dome	0.695	0.305
Combustor OP	0.545	0.416
Combustor IP	0.619	0.296

	CFD-based avg.	
Inner-Peak Profiler	Cp	Ω
Prediffuser IP exit	0.364	0.115
Prediffuser CP exit	0.313	0.199
Prediffuser OP exit	0.399	0.407
Combustor Dome	0.626	0.374
Combustor OP	0.472	0.498
Combustor IP	0.653	0.241

	CFD-based avg.	
Outer-Peak Profiler	Cp	Ω
Prediffuser IP exit	0.441	0.345
Prediffuser CP exit	0.344	0.170
Prediffuser OP exit	0.377	0.179
Combustor Dome	0.688	0.313
Combustor OP	0.600	0.364
Combustor IP	0.559	0.382

From Tables 5 and 6 follow the observations:

1. Single passage diffusion system is more robust to the variations in the prediffuser inlet profiles than a typical 3-passage diffusion system.
2. Prediffuser total pressure losses are higher in the 3-passage prediffuser along with lower static pressure recoveries in addition to having to address the variations among the three passages.

This means for the same geometric prediffuser area ratio, a single-passage diffusion system would give higher CP and lower Ω than its counter-part multi-passage diffusion system.

Table 6: Average CFD calculated CP and Ω of a typical single-passage combustor diffusion system[7]

	CFD-based avg.	
Center-Peak Profiler	CP	Ω
Prediffuser exit	0.506	0.127
Combustor Dome	0.779	0.223
Comb Outer Passage	0.529	0.455
Comb Inner Passage	0.587	0.308
	CFD-based avg.	
Inner-Peak Profiler	CP	Ω
Prediffuser exit	0.503	0.123
Combustor Dome	0.796	0.204
Comb Outer Passage	0.525	0.458
Comb Inner Passage	0.610	0.295
	CFD-based avg.	
Outer-Peak Profiler	CP	Ω
Prediffuser exit	0.505	0.127
Combustor Dome	0.772	0.235
Comb Outer Passage	0.531	0.456
Comb Inner Passage	0.575	0.328

Similar to Fig. 2, Figures 3 and 4 present the results with inner- and outer-peak profiles. There is no need to provide any further elaborations of these figures except to refer to the summary provided in Table 3. Prediffuser exit CP with inner-peak inlet profile is underpredicted by ~14%, 26% and 1% for the inner, center and outer passages, respectively; see also Figure 11. On the other hand for the outer-peak profile, CFD overpredicts CP of inner and outer by ~11% and 2%, but underpredicts the center-passage by ~14%. When averaged for all the three inlet profiles, prediffuser CP for the inner and outer passages are overpredicted by ~4% and 6%, and underpredicted by ~16% for the center passage. Even if one questions the experimental accuracy of 3-passage diffuser's CP compared to that of single passage, one couldn't ignore the obvious concern about using of the standard k-ε model for such a complex flow field.

Similar observations and discussion would lead to the following summary for the combustor dome CP. As summarized in Table 3 and pictorially shown in Figures 11 and 12, CFD overpredicts diffuser recoveries with the center- and outer-profilers by ~11% and 6%, and underpredicts by ~12% with the inner-peak profiler. Overall average overprediction for the three profiles is ~6%.

Since measuring the total pressure loss coefficient Ω in a multi-passage diffusion system is quite a challenge, there is no need to have detail discussion about CFD versus data except for the summary given in Table 3 and results presented in Figure 12.

Combustor Outer & Inner Passages CP & Ω

Figures 5, 6 and 7 show comparison between data and CFD calculated CP for the three axial stations each of the combustor outer and inner annuli (passages), respectively with the center-, inner- and outer-peak profilers. The quality of the passage CP data is expected to be better than that of the prediffuser passages. CFD reproduces the trend very well except being above or below the measured CP. Similar comparison for the total pressure loss coefficient Ω is presented in Figures 8, 9 and 10. Figures 11 and 12 show relative comparison between data (along the x-axis) and CFD calculations plotted along y-axis.

With the center-profiler, CFD overpredicts CP for both the outer and inner passages; similarly underprediction for Ω. The overall average overprediction for the combustor outer annulus CP is ~16 compared to ~6% with a single-passage diffuser[7]; the inner annulus overprediction is 7%. The overprediction values for the three axial stations of the outer annulus are ~12%, 21% and ~15%, respectively. Similarly, the underprediction values for Ω are ~15%, 14%, 16% and 15%. Results with the outer-peak profilers are similar; CP overprediction numbers are ~18%, 15%, 23% and 16%; and Ω underpredicted numbers are ~19%, 18%, 20% and 19%, respectively.

The results with the inner-peak profiler are just the opposite to those of the center- and outer-profilers. Here passage CP's are mostly underpredicted and Ω overpredicted. Ω. On the average CP of the outer and inner annuli are undepredicted by ~5% and 10%, respectively; and Ω overpredicted by 7% and 8%.

The quality of data and model correlations can be inferred from Figures 11 and 12. Each of the 12

figures show the range over which models overpredict by positive % values shown, and where underpredicted by <0 values in %. Also shown are the average values. All CP and most of total pressure loss coefficient data are correlated well, a few exceptions being the dome Ω with the inner- and outer-peak profilers.

As evident from Figures 13 and 14, the flow field is highly 3-dimensional caused partially by the diffuser geometry, fuel nozzles and mixer. Having the ability to calculate the flow field details, even when qualitatively accurate, gives tremendous insight to the design's critical features.

SUMMARY

A full-scale 3-passage combustor diffusion system has been analyzed with Fluent code using its grid generation package, the standard k-ε model and the nonequilibrium wall function. Inlet conditions were calculated using through the rig inlet, the three profilers that provide center-, inner- and outer-peak profiles at the prediffuser inlet. Predictions were compared with data from a full-scale annular diffuser rig. In general, CFD gave good agreement with data as shown in the following table.

Errors (Data-CFD)/CFD) for

	Prediffuser exit			Dome		
	Average	STDEV	RMSE	Average	STDEV	RMSE
Center-Peak Profiler						
2D 1P[6]	-0.2%	1.4%	1.4%	25.0%	2.2%	25.1%
3D 1P[7]	-2.2%	0.5%	2.2%	7.6%	1.5%	7.7%
3D 3P	-6.4%	3.5%	14.1%	12.4%	1.8%	12.2%
Inner-Peak Profiler						
2D 1P[6]	2.2%	1.2%	2.3%	26.7%	3.9%	24.4%
3D 1P[7]	0.3%	1.1%	1.1%	6.0%	2.7%	6.6%
3D 3P	15.4%	3.1%	18.3%	25.5%	5.3%	26.0%
Outer-Peak Profiler						
2D 1P[6]	11.0%	1.5%	8.9%	32.1%	2.2%	25.8%
3D 1P[7]	11.1%	1.0%	11.1%	15.8%	1.6%	15.9%
3D 3P	1.3%	2.6%	10.6%	14.1%	1.5%	14.0%
	Outer Passage			Inner Passage		
Center-Peak Profiler						
2D 1P[6]	-11.0%	3.2%	11.8%	-11.4%	3.9%	12.1%
3D 1P[7]	-6.3%	2.9%	6.9%	-9.1%	3.0%	9.6%
3D 3P	-15.9%	9.3%	14.6%	-7.1%	4.0%	8.3%
Inner-Peak Profiler						
2D 1P[6]	-6.1%	2.4%	6.3%	-14.1%	11.3%	16.2%
3D 1P[7]	-3.4%	2.1%	3.9%	-8.3%	3.2%	8.5%
3D 3P	4.6%	3.4%	4.7%	9.9%	4.9%	11.2%
Outer-Peak Profiler						
2D 1P[6]	-1.2%	3.7%	3.4%	5.8%	4.6%	5.9%
3D 1P[7]	7.5%	2.1%	7.8%	5.3%	2.4%	5.9%
3D 3P	-18.2%	9.8%	16.1%	3.9%	1.3%	4.3%

2-D[6] and 3-D calculations[7] for single passage diffuser are compared with the 3-passage diffuser in regard to errors (defined as 1-CFD/Data), average and its standard deviation, and root-mean square error for CP of the prediffuser, dome, outer and inner combustor annuli (passages). Further model validation work is planned 11-14 to extend the range of application as well as improving accuracy.

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Table 1: Typical 3-D Combustor Diffuser Runs

Run #	Comments
3	Center-peaked inlet profile.
4	Inner-peaked inlet profile.
5	Outer-peaked inlet profile.
10	Repeat of Run #3
6	Center-peaked inlet profile w/o cowling
7	Center-peaked inlet profile w/ reduced cowling open area
8	Opposed ring inlet profiler to compare w/ Run 3
9	Center-peaked profile; increased area shorter diffuser
11	Center-peaked profile; increased nozzle dia to 1.9", slightly modified cowling.
12	Run 11 w/ 20% increased nozzle blockage.

Table 2: Average errors between CFD and data compared for 2-D and 3-D simulations.

	Prediffuser exit		Dome		Outer Passage		Inner Passage	
	CP	Ω	CP	Ω	CP	Ω	CP	Ω
Run 3/2D	1.37%	11.53%	25.07%	151.87%	14.50%	15.08%	10.57%	15.57%
Run 3/3D	2.21%	8.44%	7.74%	50.22%	6.97%	6.64%	10.75%	11.76%
Run 4/2D	2.00%	18.87%	21.62%	139.88%	7.14%	7.65%	11.53%	11.64%
Run 4/3D	1.09%	8.69%	6.56%	47.09%	3.77%	3.64%	9.51%	12.31%
Run 5/2D	8.62%	40.44%	24.94%	484.87%	3.80%	4.89%	6.82%	15.86%
Run 5/3D	11.14%	98.69%	15.86%	321.11%	7.84%	12.07%	5.00%	14.98%

Table 3: Average errors (1-CFD/Data) in CP and Ω of typical 3-passage diffusion system.

CP	Prediffuser exit inner, center & outer			Dome	OP3.5	IP3.5	OP3.6	IP3.6	OP3.8	IP3.8
Avg Error, CenterPeak	-14.45%	8.95%	-17.44%	12.43%	-11.60%	-9.18%	-21.43%	-7.19%	-14.77%	-5.00%
Avg Error, InnerPeak	13.77%	25.85%	1.26%	25.49%	7.57%	7.24%	0.90%	11.57%	5.37%	11.02%
Avg Error, OuterPeak	-11.21%	13.56%	-2.46%	14.00%	-15.07%	2.26%	-23.04%	4.17%	-16.45%	5.30%
Avg Error	-3.96%	16.12%	-6.21%	17.31%	-6.36%	0.11%	-14.53%	2.85%	-8.62%	3.77%
Overall avg	0.05%	σ	13.41%	Max	25.85%	Min	-23.04%	RMS	13.18%	
Ω	Prediffuser exit inner, center & outer			Dome	OP3.5	IP3.5	OP3.6	IP3.6	OP3.8	IP3.8
Avg Error, CenterPeak	11.24%	-136.17%	29.18%	-49.82%	13.55%	27.30%	16.16%	22.62%	14.72%	14.58%
Avg Error, InnerPeak	313.22%	-67.92%	2.88%	-187.56%	-8.16%	11.83%	-5.23%	-6.95%	-6.28%	-30.36%
Avg Error, OuterPeak	-25.85%	-517.07%	43.88%	-63.60%	18.41%	2.85%	19.97%	-1.63%	18.82%	-6.54%
Avg Error	99.54%	-240.38%	25.31%	-100.33%	7.94%	13.99%	10.30%	4.68%	9.09%	-7.44%
Overall avg	-17.73%	σ	121.63%	Max	313.22%	Min	-517.07%	RMS	120.89%	

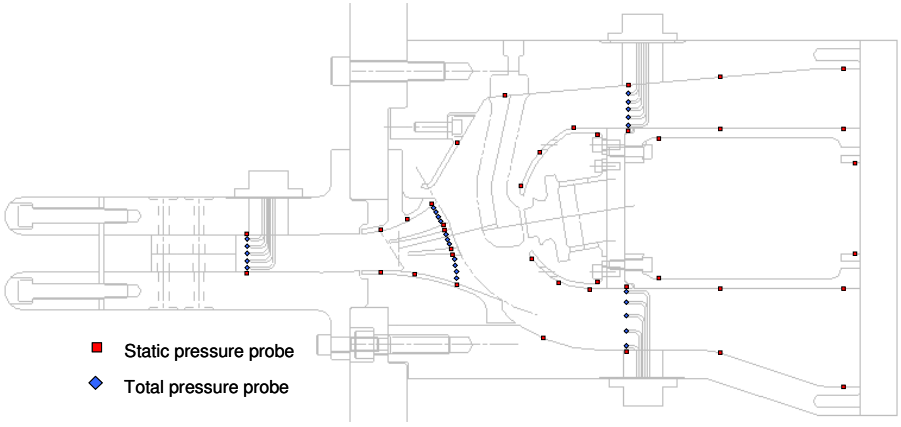
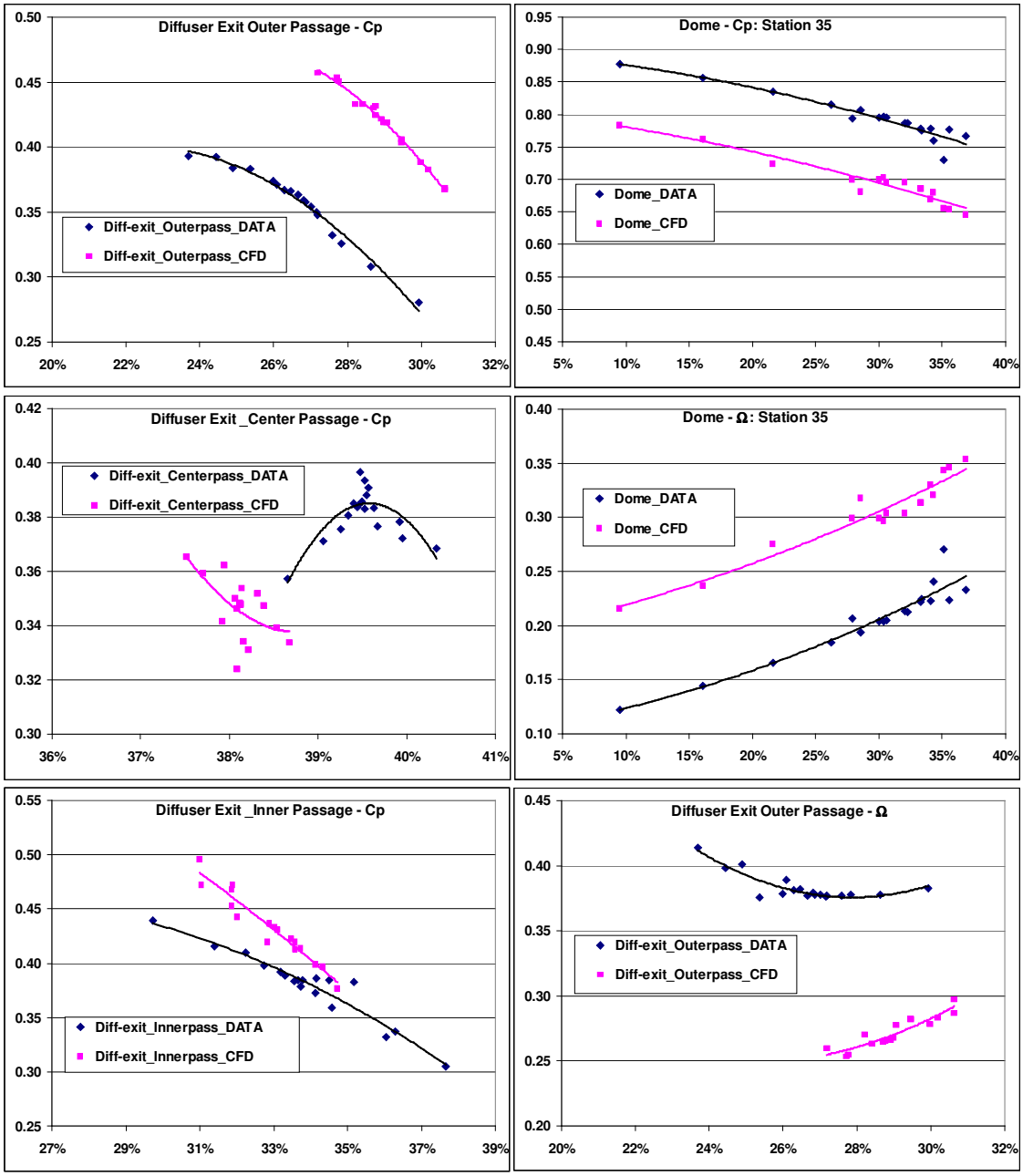


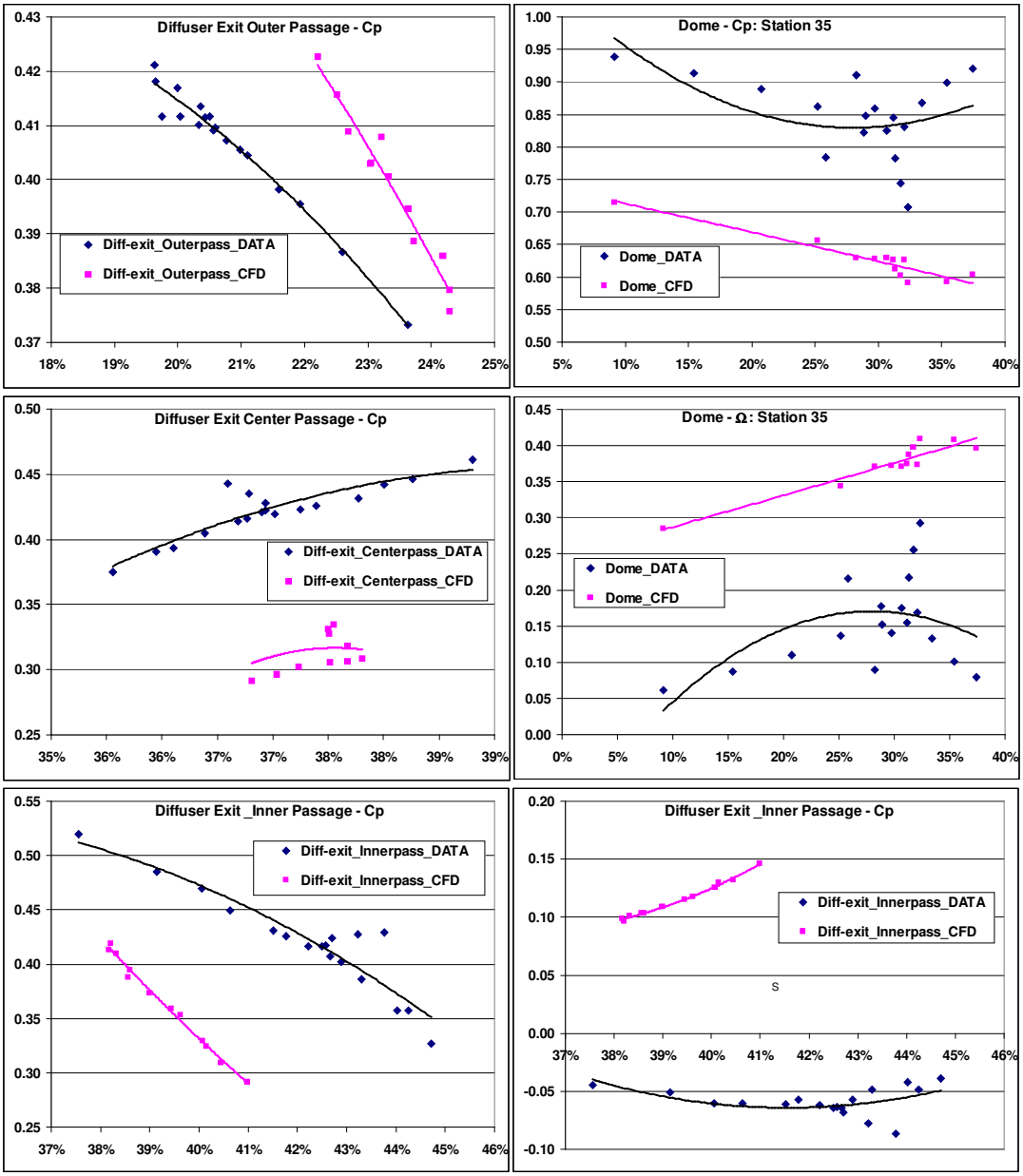
Figure 1: A typical 3-passage diffusion system analyzed.



Diffuser Exit Cp

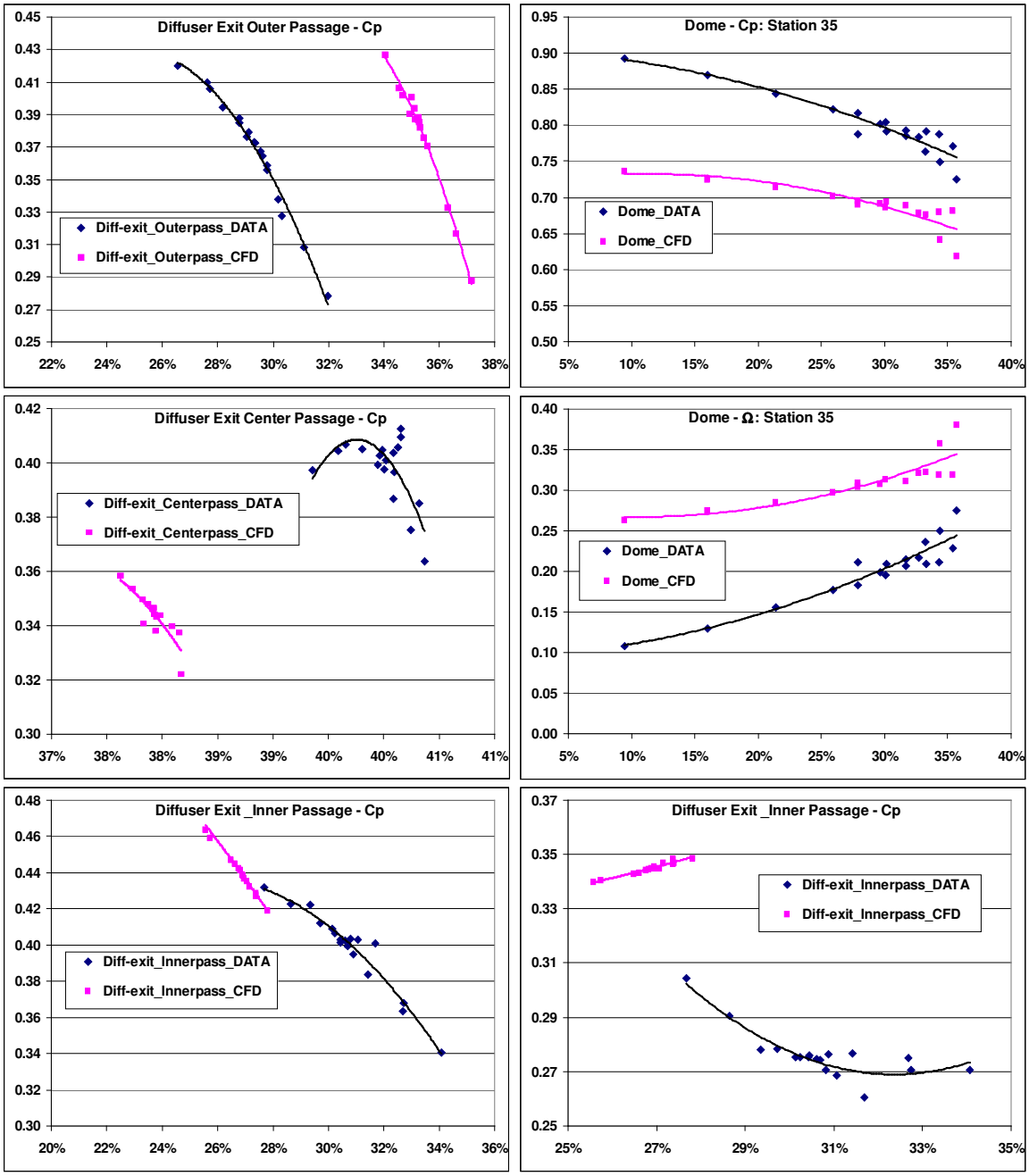
Dome Cp and Ω, Diffuser Exit Ω

Figure 2: Comparison between measured and predicted Cp and Ω for diffuser exit and dome passages with center-peak profiler. Flow split plotted along x-axis; Cp or Ω along y-axis.



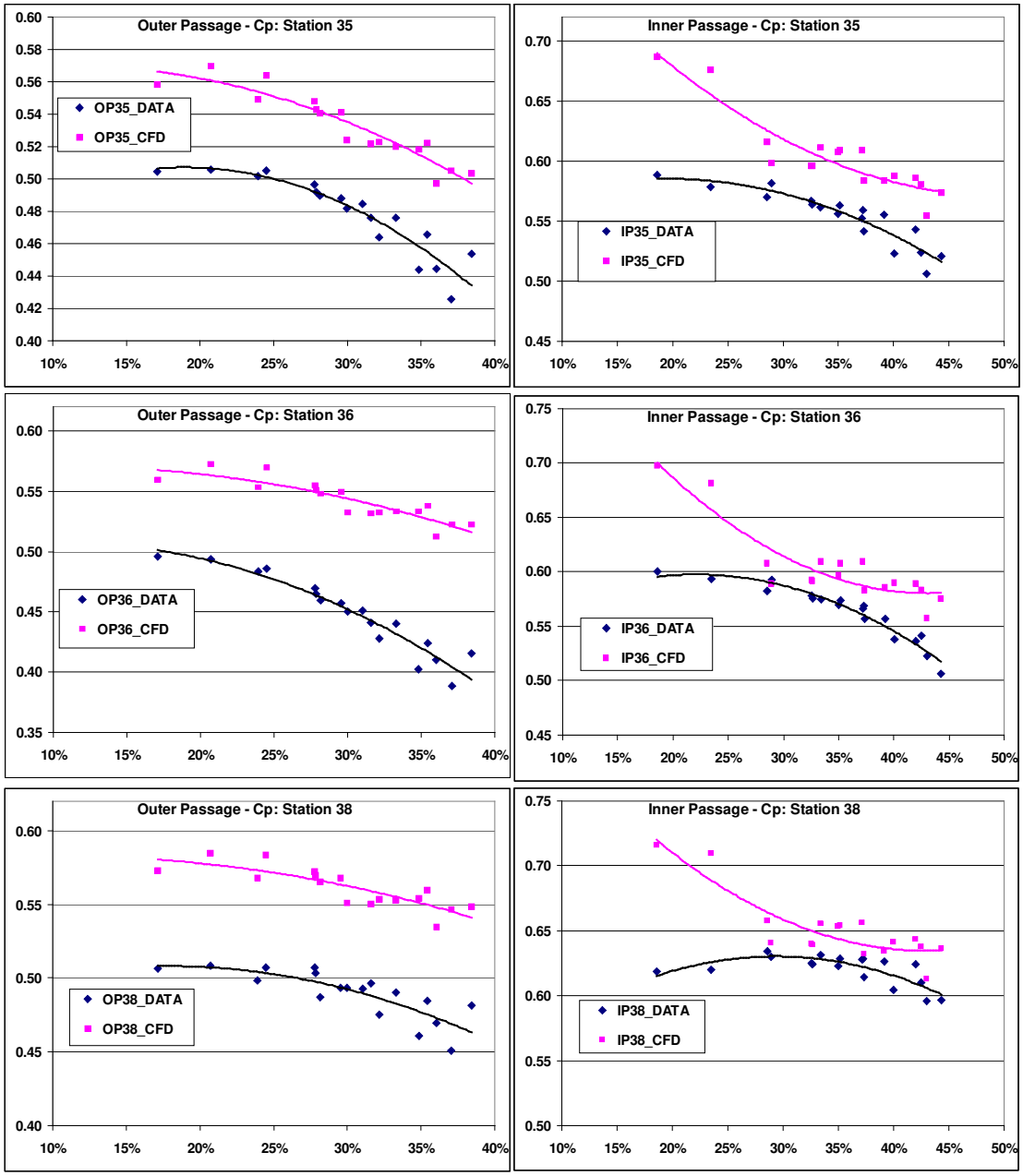
Diffuser Exit Cp Dome Cp and Ω, Diffuser Exit Ω

Figure 3: Comparison between measured and predicted Cp and Ω for diffuser exit and dome passages with inner-peak profiler. Flow split plotted along x-axis; Cp or Ω along y-axis.



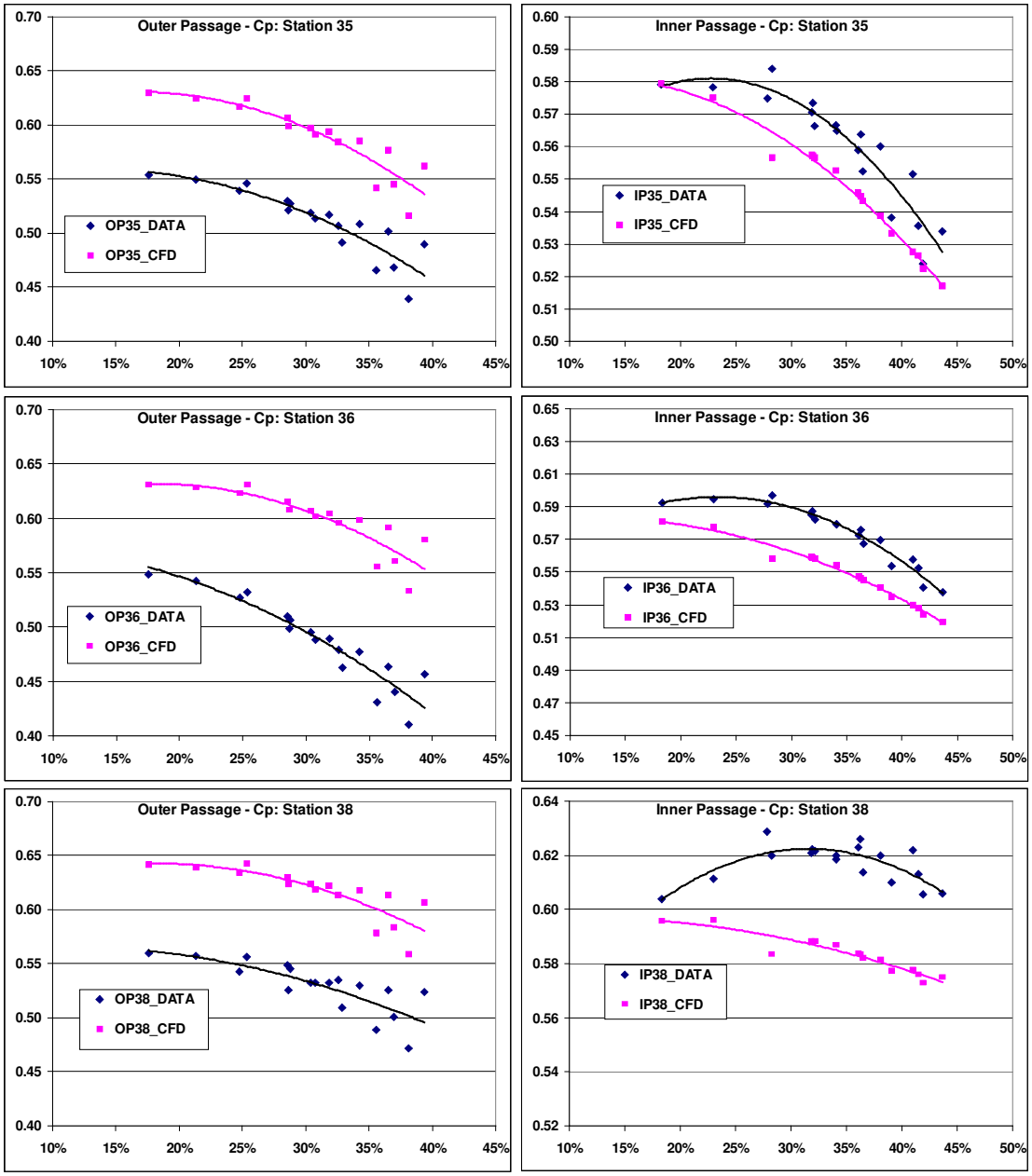
Diffuser Exit Cp Dome Cp and Ω, Diffuser Exit Ω

Figure 4: Comparison between measured and predicted Cp and Ω for diffuser exit and dome passages with outer-peak profiler. Flow split plotted along x-axis; Cp or Ω along y-axis.



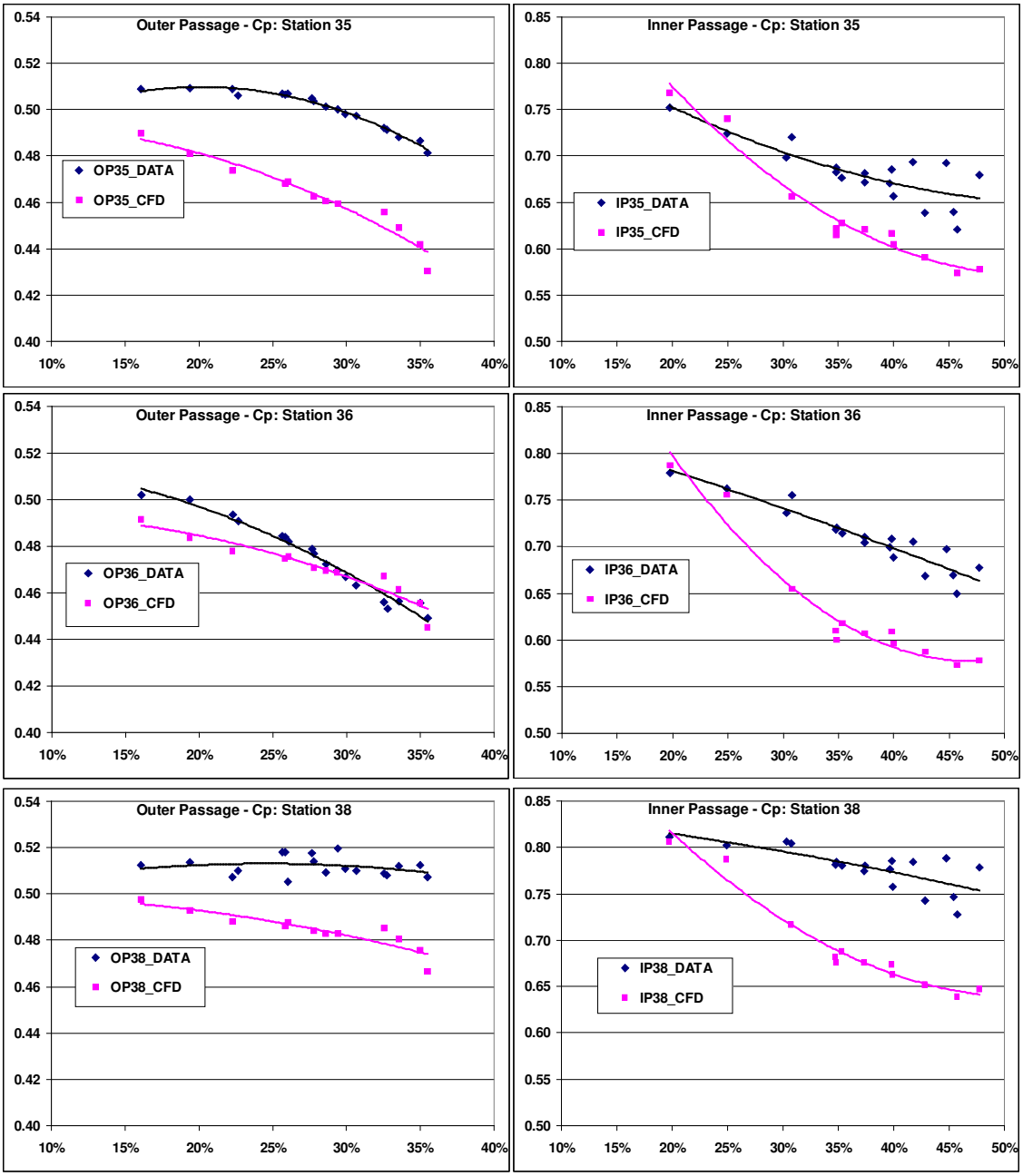
Outer Passage Cp Inner Passage Cp

Figure 5: Comparison between measured and predicted Cp for outer and inner passages with center-peak profiler. Flow split plotted along x-axis; Cp along y-axis.



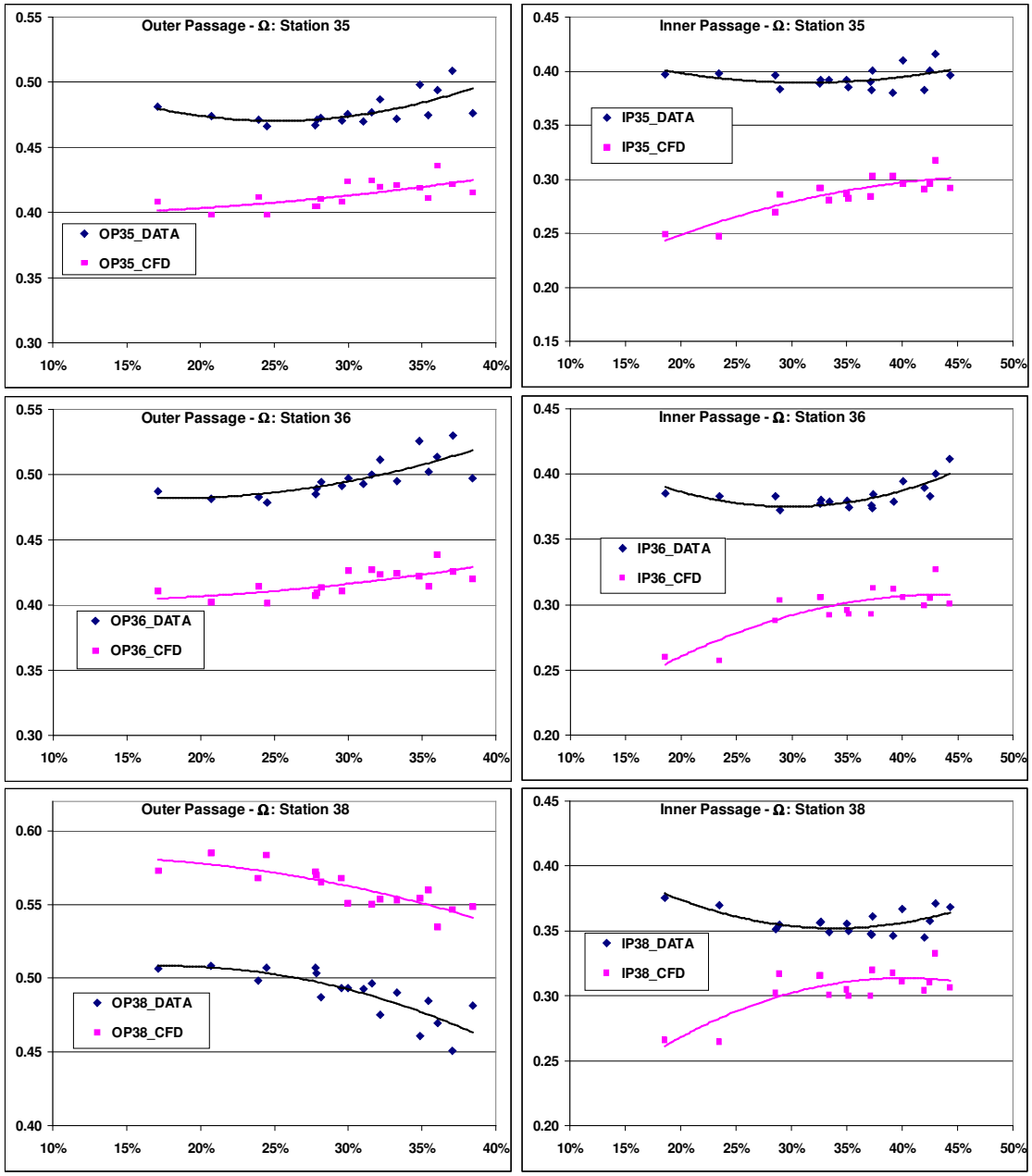
Outer Passage Cp Inner Passage Cp

Figure 6: Comparison between measured and predicted Cp for outer and inner passages with inner-peak profiler. Flow split plotted along x-axis; Cp along y-axis.



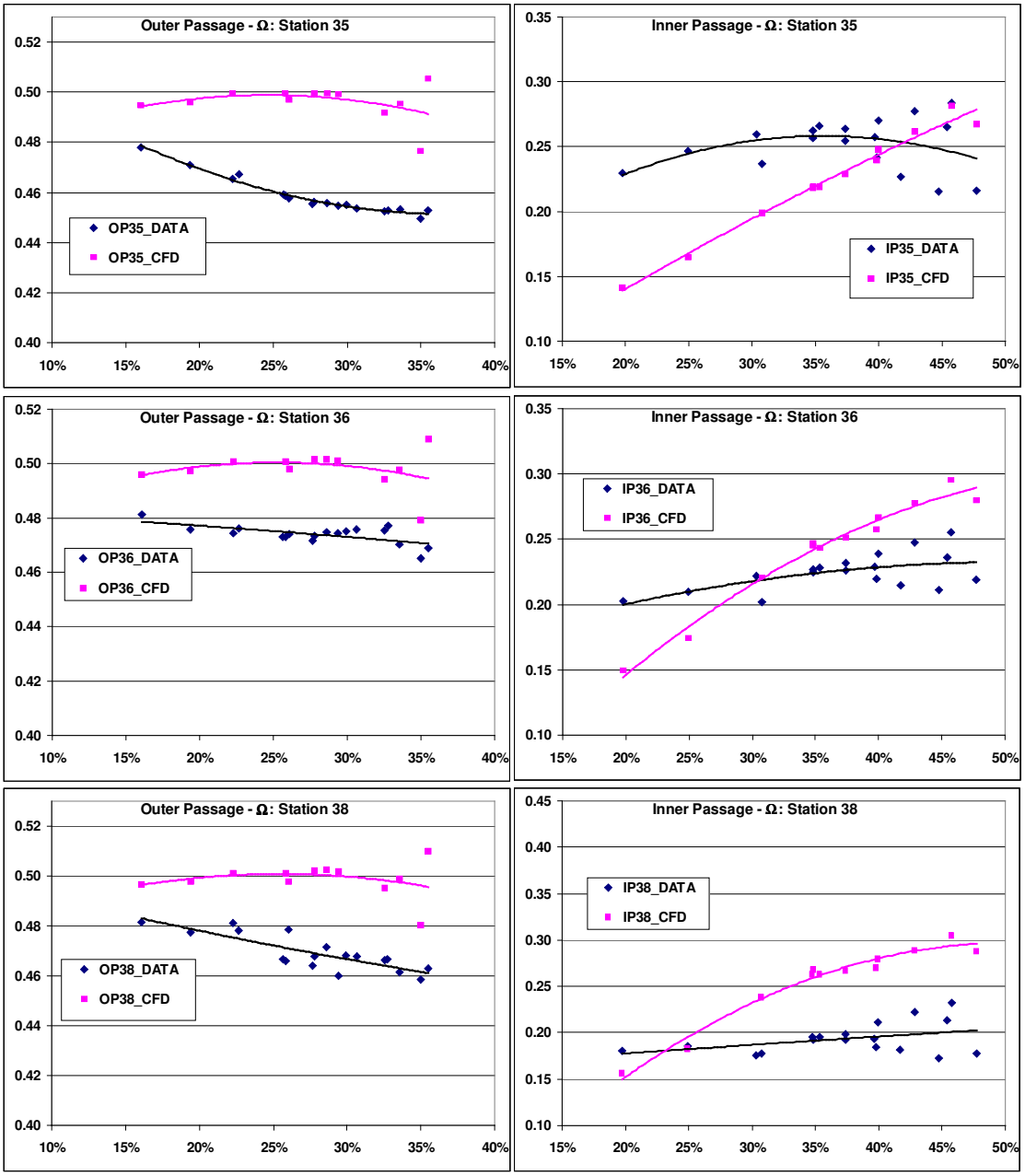
Outer Passage Cp Inner Passage Cp

Figure 7: Comparison between measured and predicted Cp for outer and inner passages with outer-peak profiler. Flow split plotted along x-axis; Cp along y-axis.



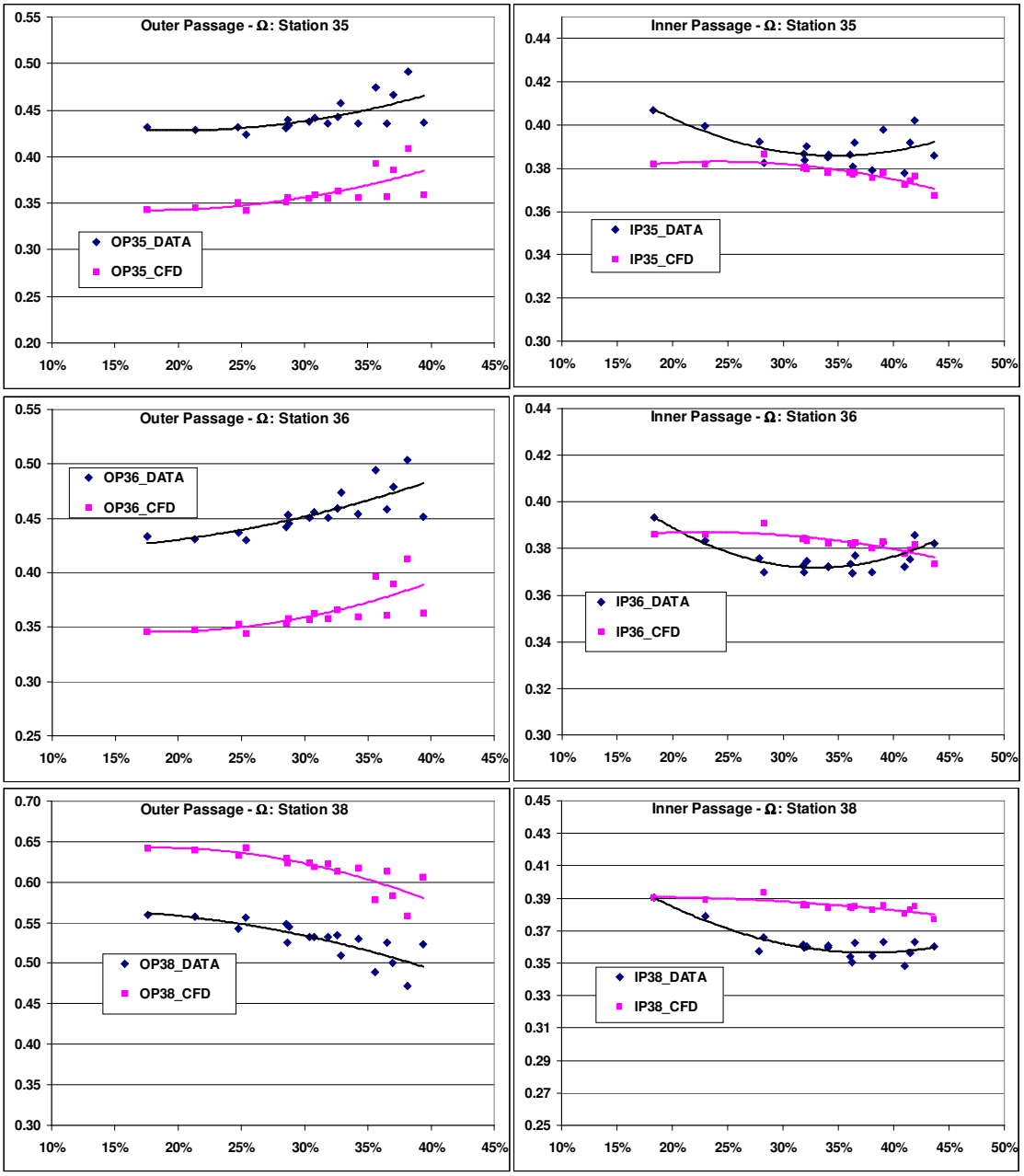
Outer Passage Ω Inner Passage Ω

Figure 8: Comparison between measured and predicted Ω for outer and inner passages with center-peak profiler. Flow split plotted along x-axis; Ω along y-axis.



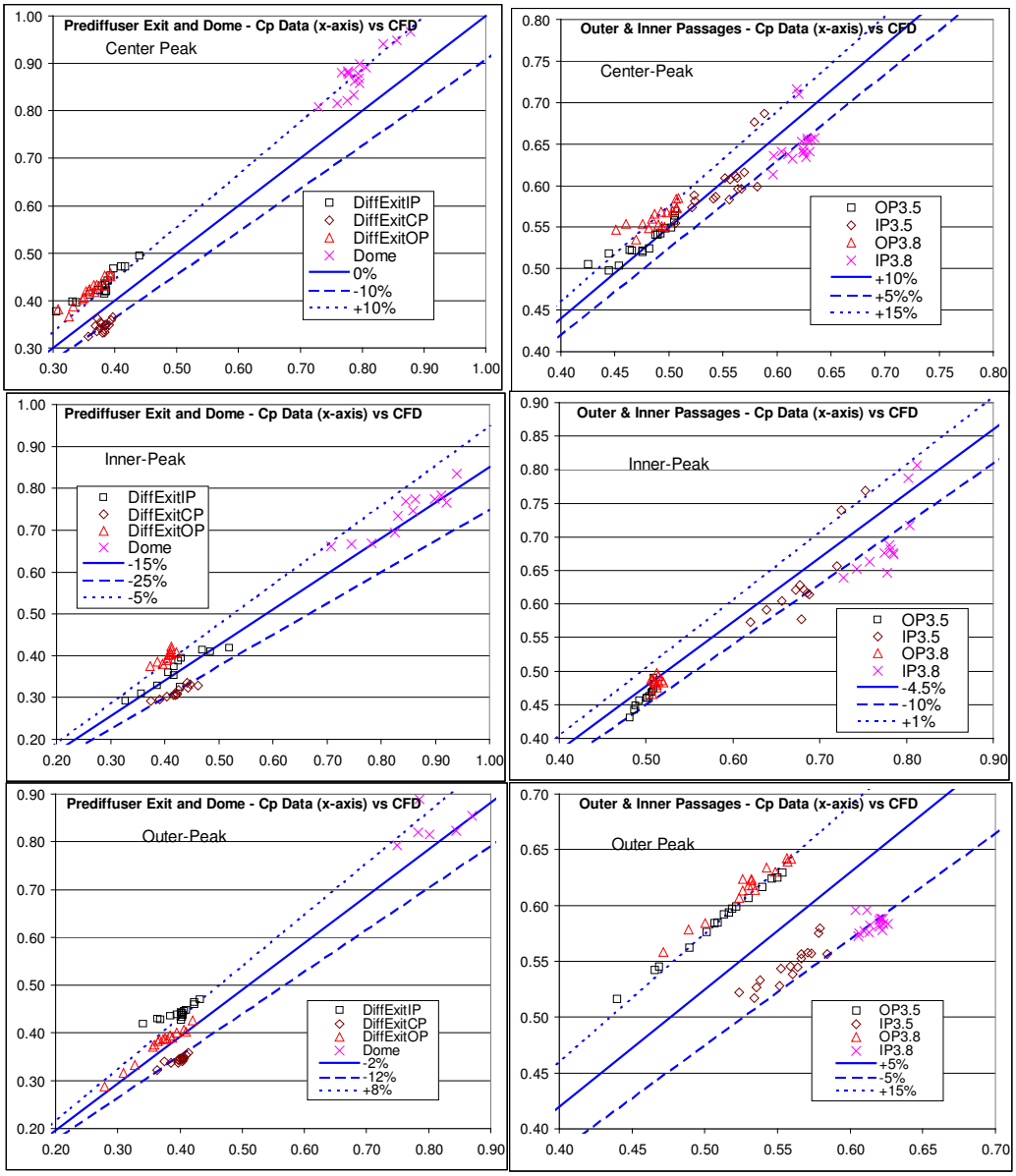
Outer Passage Ω Inner Passage Ω

Figure 9: Comparison between measured and predicted Ω for outer and inner passages with inner-peak profiler. Flow split plotted along x-axis; Ω along y-axis.



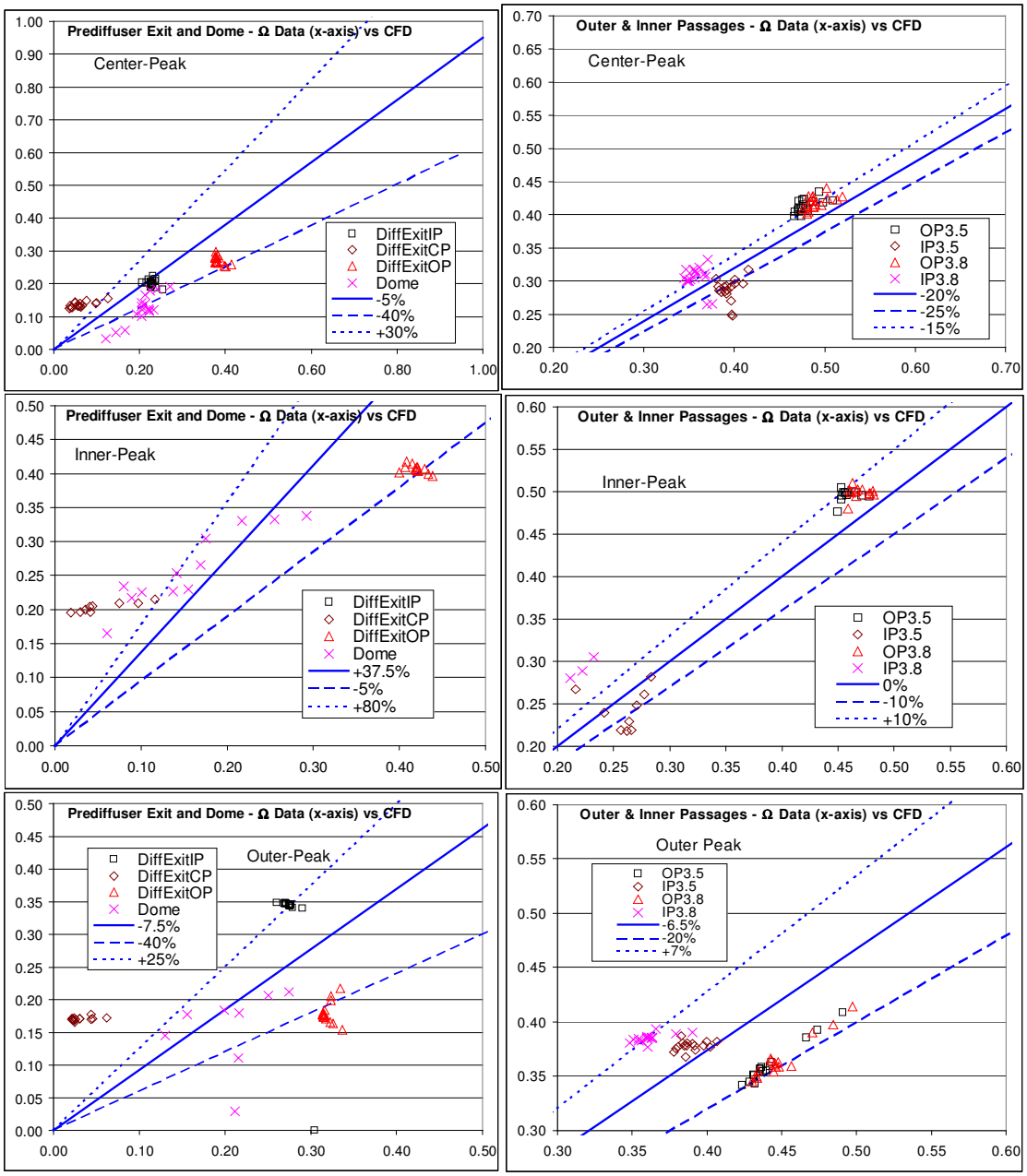
Outer Passage Ω Inner Passage Ω

Figure 10: Comparison between measured and predicted Ω for outer and inner passages with outer-peak profiler. Flow split plotted along x-axis; Ω along y-axis.



CP	Prediffuser exit inner, center & outer			Dome	OP3.5	IP3.5	OP3.6	IP3.6	OP3.8	IP3.8	Profiler
Avg Error	-14.45%	8.95%	-17.44%	-10.54%	-11.60%	-9.18%	-21.43%	-7.19%	-14.77%	-5.00%	Center
Avg Error	13.77%	25.85%	1.26%	12.26%	7.57%	7.24%	0.90%	11.57%	5.37%	11.02%	Inner
Avg Error	-11.21%	13.56%	-2.46%	-6.35%	-15.07%	2.26%	-23.04%	4.17%	-16.45%	5.30%	Outer
Overall avg	-1.84%	σ	12.34%	Max	25.85%	Min	-23.04%				

Figure 11: Comparison between measured and predicted Cp for center- (top), inner- (middle) and outer-peak (bottom) profilers. Data plotted along x-axis; CFD along y-axis.



Ω	Prediffuser exit inner, center & outer			Dome	OP3.5	IP3.5	OP3.6	IP3.6	OP3.8	IP3.8	Profiler
Avg Error	11.24%	-136.17%	29.18%	42.45%	13.55%	27.30%	16.16%	22.62%	14.72%	14.58%	Center
Avg Error	313.22%	-67.92%	2.88%	-88.13%	-8.16%	11.83%	-5.23%	-6.95%	-6.28%	-30.36%	Inner
Avg Error	-25.85%	-517.07%	43.88%	26.20%	18.41%	2.85%	19.97%	-1.63%	18.82%	-6.54%	Outer
Overall avg	-8.35%	σ	118.16%	Max	313.22%	Min	-517.07%				

Figure 12: Comparison between measured and predicted Ω for center- (top), inner- (middle) and outer-peak (bottom) profilers. Data plotted along x-axis; CFD along y-axis.

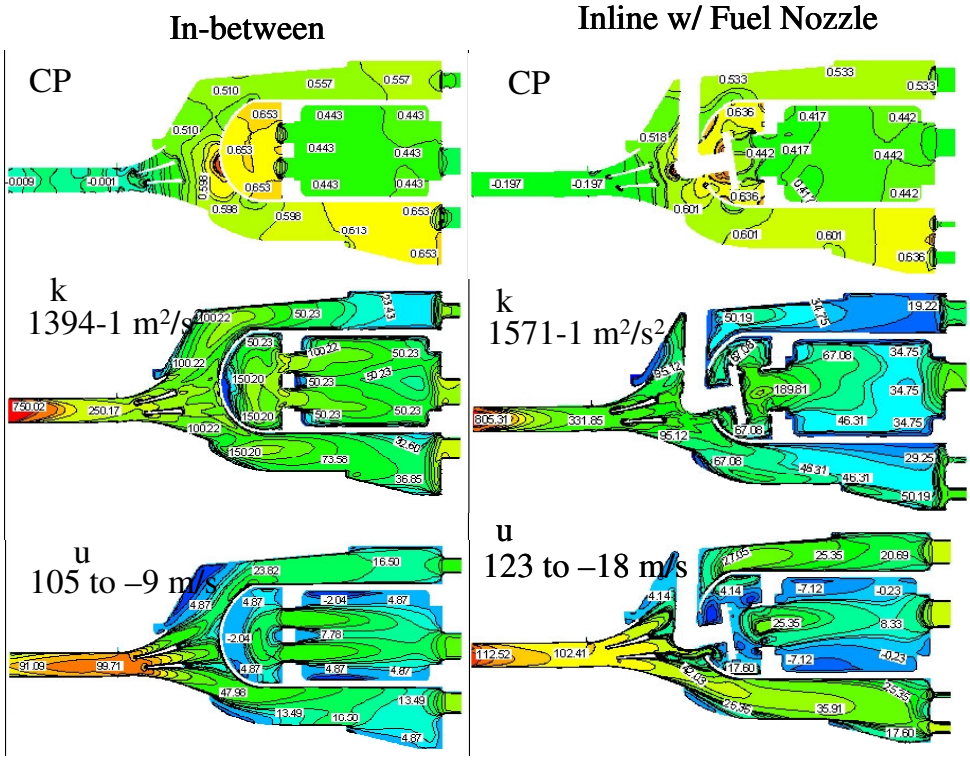


Figure 13: Typical contours of CP, turbulence kinetic energy and axial velocity In-between and in-line with fuel nozzle stem.

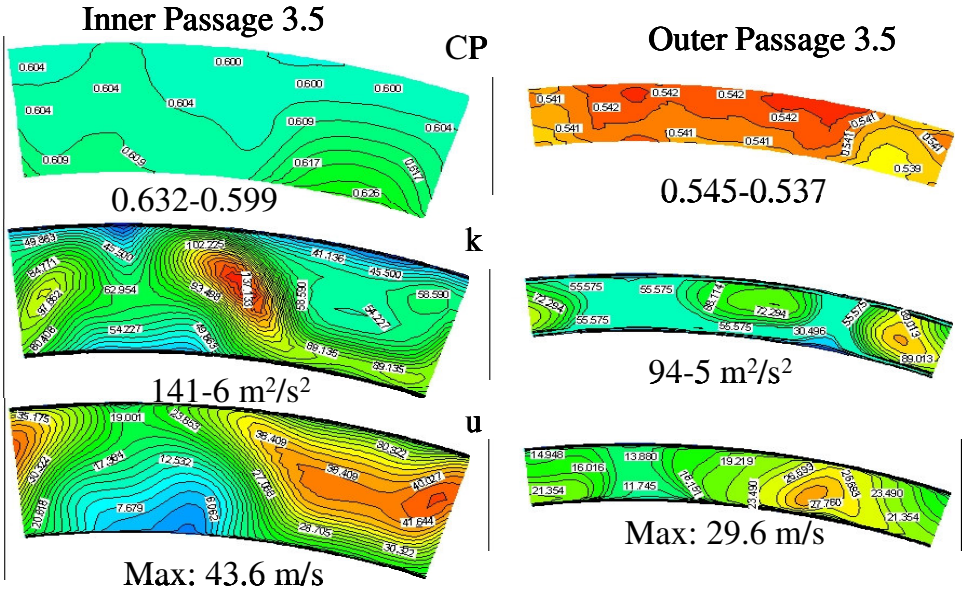


Figure 14: Typical contours of CP, turbulence kinetic energy and axial velocity at 3.5 axial planes of the inner and outer passages of the combustor diffusion system.